朋友 Space (Pengyou Space): A Case Study on Developing a Hyper-Local Social Media Platform to Alleviate Student Social Isolation on University Campuses

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Report date: 14 October 2026

Abstract

Context. Informal, unplanned face-to-face interaction — "hanging out" — is a recognised contributor to students' sense of belonging and academic integration, yet many university students report difficulty arranging it. **Problem.** General-purpose social media is poorly suited to campus-scale offline discovery: it is not scoped to the campus, is not organised around place and time, and its passive-consumption loop can deepen rather than relieve isolation. **Approach.** This paper presents 朋友 Space (Pengyou Space), a hyper-local, university-scoped social discovery platform deployed at Nantong University, and evaluates its use empirically rather than describing its architecture. **Method.** A single-group, self-report instrument of 14 items — three five-point scales, three multi-select batteries and three open questions — was distributed through an in-app banner and a campus WeChat group between 1 and 14 October 2026; 150 enrolled students who had used the platform at least once responded, and every statistic reported here was computed from those raw records with its denominator stated. **Key results.** Before adoption, 77.3% of respondents (n = 116/150) rated finding hangouts difficult or very difficult, led by not knowing where people gather (45.3%) and approach anxiety (42.7%). After adoption, 86.7% (n = 130/150) reported discovery much or somewhat easier, and the most frequently reported benefits correspond to the main challenges respondents had named: discovery (53.3%), interest matching (52.7%), event finding (49.3%) and reduced awkwardness

(48.0%). Satisfaction reached 82.0% (n = 123/150) and at-least-weekly use 82.0% (n = 123/150); satisfaction rose with campus tenure from 79.5% among residents of under six months to 90.6% among residents of more than two years, while all negative satisfaction responses fell in the newest cohort. Pre-adoption difficulty was high at every degree level (73.0–81.8%), indicating a structural rather than a transitional gap. **Contribution.** The paper contributes a reproducible campus-discovery instrument with released frequency tables, a documented citation audit, and first-party evidence that platform use is *associated* with large reported reductions in discovery difficulty — while bounding all causal claims, since a single-group self-report design cannot separate the platform's effect from time, novelty or self-selection.

Keywords: hyper-local social platform; student social isolation; campus discovery gap; perceived belonging; self-report survey evaluation; university-scoped social media

1. Introduction

1.1 Social connection and student outcomes

The need to belong is among the most stable findings in motivational psychology. Baumeister and Leary (1995) argued that human beings require frequent, affectively pleasant interactions with a small set of stable persons — not as a luxury or a personality quirk, but as a fundamental motivation with measurable costs when unmet. For university students, that requirement carries institutional consequences. Tinto's (1993) model of student departure made social integration a first-class determinant of persistence alongside academic integration: students who fail to weave themselves into the informal fabric of campus life are measurably more likely to leave it. Ellison et al. (2007) showed that the digital layer of that fabric matters — university-anchored online network sites convert latent ties into bridging social capital, particularly for students who begin college lonely — and Steinfield et al. (2008) extended

the finding longitudinally, linking network-site use to social capital and self-esteem in a reciprocal pattern.

What these traditions share is an emphasis on the *informal*: the shared desk in a study hall, the impromptu dinner, the walk to a sports ground after class. Formal student organisations — clubs, societies, class groups — have discovery channels by construction; timetables, rosters and recruitment fairs make them visible. Informal co-presence has no such channel. It is arranged, when it happens at all, through improvised coordination between people who already know one another. The present study takes that improvised layer as its object.

Against this background, student social isolation has become a salient public-health concern. Using a nationally representative sample of U.S. young adults, Primack et al. (2017) found perceived social isolation concentrated in exactly the age band that populates university campuses, and associated with patterns of social media use of the kind students live inside daily. Their setting is not Chinese higher education, and this paper does not import their effect sizes; the pattern nonetheless motivates the engineering question this paper pursues: can the discovery layer of campus sociability — knowing who is free, where people gather, and how to join without embarrassment — be redesigned as a small, purpose-built software system, and can that redesign be evaluated with honest, first-party evidence?

1.2 The campus discovery gap

Most campuses already contain the raw material of friendship: hundreds of students with compatible schedules, overlapping interests and close physical proximity. What is missing, students report, is a discovery channel. In the survey that underlies this paper, 116 of 150 respondents (77.3%) rated finding informal hangouts difficult or very difficult before they ever used 朋友 Space; the two most-mentioned barriers were not knowing where people usually hang out (45.3% of respondents) and feeling shy or awkward approaching others (42.7%) — a coordination problem and an emotional problem, not a motivation problem. Desire was not lacking: 86.7% of the same sample wanted to hang out at least once a month (Section 4.2).

The existing channels were analog or diffuse. Respondents named asking friends or classmates (50.0%), generic social media (46.0%), campus posters (31.3%), clubs and societies (30.7%) and simply walking around campus (20.0%) — the full battery is reported in Section 4.4. None of these answers the question a lonely Tuesday afternoon actually poses: *who near me is doing something I could join in the next hour?* This paper calls that unanswered question the **campus discovery gap**, and treats it as a design object rather than a personality trait. The framing matters for what follows: if difficulty is a discovery problem, the remedy is an information and affordance system, not a campaign of encouragement.

1.3 Why general-purpose social media fails

General-purpose platforms are the first thing students reach for, and they fail the discovery gap in three specific ways. First, they are not *scoped*. A mainstream feed mixes classmates with strangers and alumni with nearby residents, and the campus boundary — the fact that makes approach feel safe — is absent. Users of 朋友 Space in this study named the campus-scoped structure as their single most-liked property: "feels safer and more structured than random social media" was the most frequent open-text theme (17 of 129 non-empty responses; Section 4.10). Second, general platforms are not *place-forward*. Nothing in a global feed is organised around the physical spot and time at which co-presence could occur. Wellman (2001) observed early on that community in the networked era persists as *personalized networks* — ties that cross places rather than places themselves — but a campus app must do the inverse operation: re-anchor those crossing ties to concrete benches, courts and cafés, because on a campus the place *is* the product.

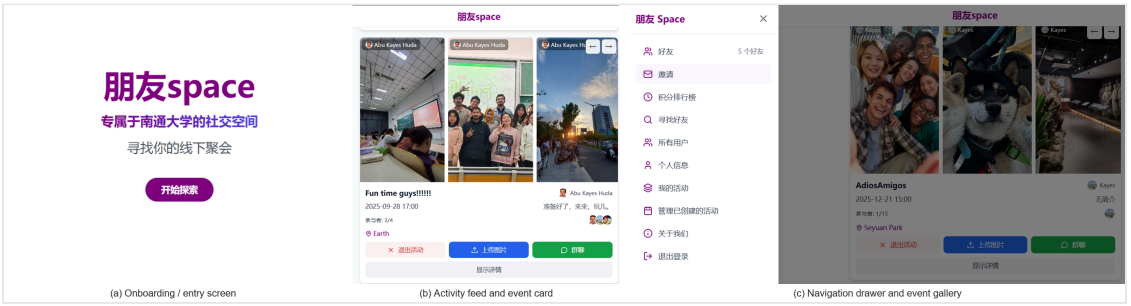
Third, and most consequentially, general-purpose feeds can substitute parasocial consumption for contact. The evidence that well-being effects are design-dependent rather than medium-dependent is now broad: Burke and Kraut (2016) found that Facebook use paid off in well-being only when communication was direct and matched tie strength; Valkenburg and Peter (2007) showed that online communication benefits adolescent well-being when it *stimulates* offline interaction rather than displacing it; and Przybylski et al. (2013) identified fear of missing out as the motivational engine of the compulsive, passive checking loop that displaces contact. A feed optimised for

engagement time converts isolation into more scrolling. Primack et al. (2017) report the association between social media use and perceived isolation in exactly this population. The synthesis is uncomfortable but clear: the same technology that could bridge the discovery gap is, as ordinarily designed, one of the gap's producers.

1.4 The case for a scoped, discovery-first platform

If the failure modes are scoping, place and loop design, the remedy is a platform whose every surface answers a discover–join–meet loop. 朋友 Space implements that idea (Figure 1). It is a university-scoped social discovery platform: membership of the host university is the precondition of visibility; the primary navigation object is a time-and-place card — an event or hangout with visible participants and capacity — rather than a global content feed; joining is a single explicit action with social context that lowers the approach cost; and the terminal state of every feature is an attendance record in the physical world, not a session length on the screen. The system is deliberately small — a React and TypeScript web application on a managed backend — because the claim being made is a design claim, not an infrastructure claim: the discovery layer of campus sociability is cheap to build, and what remains is to evaluate it honestly.

That honest evaluation is the second half of this paper. A single-group, self-report survey of 150 student users captures the pre-adoption difficulty, post-adoption ease, benefit structure, satisfaction and engagement patterns that the design predicts. The evidence is associative, not causal — Section 5.5 says so repeatedly — but it is dense, internally consistent, and traceable to raw response records.



(a) Onboarding / entry screen (b) Activity feed and event card (c) Navigation drawer and event gallery

Figure 1. The 朋友 Space interface as deployed: (a) onboarding entry screen; (b) activity feed — a time-and-place card with participant avatars, join/leave actions, photo upload and group chat; (c) navigation drawer with friends, invitations, a points leaderboard and an event gallery. *Source: author's software screenshots (2026).*

1.5 Contribution and research questions

This paper makes two contributions. First, it documents a complete, reproducible case of a hyper-local campus platform: the system under study, its instrumented evaluation, and the full 14-item survey battery are released together (Appendix A), so a second campus can repeat the procedure with its own deployment. Second, it reports a first-party empirical evaluation of that deployment: n = 150, fourteen items, eighteen figures and eleven tables in the body, every statistic computed from raw response records with its denominator stated. Four research questions structure the evaluation. **RQ1.** How difficult do students report finding informal hangouts before adopting the platform, and what specifically do they name as barriers? **RQ2.** After adoption, what changes do they report in the ease of discovery, matching and joining, and how do reported benefits correspond to the named challenges? **RQ3.** Which platform properties do users themselves identify as valuable, and which improvements do they request? **RQ4.** What engagement and satisfaction patterns characterise continued use, and how do these vary by year of study and campus tenure? Because the study is a single-group post-hoc survey, the questions are framed around *reports* and *associations*, not causal effects.

1.6 Paper organisation

Section 2 positions the work within five literatures and states the gap. Section 3 describes methodology: the system under study (Section 3.1), research design (Section 3.2), instrument (Section 3.3), participants and the data provenance declaration (Section 3.4), procedure (Section 3.5), analysis plan (Section 3.6) and ethics (Section 3.7). Section 4 presents results — values only, no interpretation — across eleven subsections covering participant profile, desired social frequency, pre-adoption difficulty and challenges, prior discovery methods, post-adoption ease, reported benefits, satisfaction and engagement, the before-versus-after comparison, cross-tabulations by tenure and year of

study, and the open-text themes. Section 5 interprets the findings: overall interpretation, mapping to design decisions, relation to the wider literature, implications for practice, limitations, threats to validity and future work. Section 6 concludes. References follow, then Appendix A (survey instrument), Appendix B (full frequency tables) and Appendix C (citation verification audit).

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2. Related Work

2.1 Place, community, and technology

Networked-community research has spent three decades debating whether cyberspace dissolves place. Wellman's (2001) synthesis argued that it does: community has shifted from door-to-door groups to *personalized networks* — ties that travel with the person, leaving geography behind. The Netville study complicated that picture. Hampton and Wellman (2003) followed residents of a wired Canadian suburb and found that an electronic network *scoped to the neighbourhood* increased door-to-door awareness, local contact and neighbouring — the network amplified the place rather than escaping it. The lesson is conditional: a network amplifies its scope. A globally scoped feed amplifies global ties; a campus-scoped network can amplify the campus. 朋友 Space is built on that condition. Its boundary rule — university membership as the precondition of visibility — is the Netville mechanism translated into a database policy, and its feed is organised around spots and times so that the amplified ties land on concrete benches, courts and cafés.

2.2 Social capital, belonging, and student outcomes

The second strand concerns what campus connections are *for*. Ellison et al. (2007) distinguished bridging social capital — loose, heterogeneous ties that carry new information and novel contacts — from bonding capital, and found site use associated with bridging gains, with the largest effects among students who reported high loneliness at entry. Steinfield et al. (2008) tracked the same mechanisms longitudinally and found reciprocal relationships between network-site use, social capital and self-esteem. Baumeister and Leary (1995)

supply the motivational substrate: these are not frivolous ties. Tinto (1993) supplies the institutional stakes: social integration predicts persistence. The practical implication of the four papers read together is that the informal layer of campus life is load-bearing for student outcomes, and that digital tools which convert latent ties into co-presence can plausibly move it. The caution read together is equally clear: every one of these findings is correlational, which sets the inferential standard this paper adopts for its own results.

2.3 Social media and well-being: the design dependency

The third strand explains why "social media" is the wrong unit of analysis. Burke and Kraut (2016) found that the well-being return of Facebook use depended on communication type: direct, dyadic messages to strong ties helped; passive consumption and broadcast to weak ties did not. Valkenburg and Peter (2007) tested the stimulation versus displacement hypotheses in a panel of adolescents and found support for stimulation — online communication improved well-being when it stimulated offline interaction — concentrated in the users who needed it most. Przybylski et al. (2013) named the engine of the opposite loop: fear of missing out, a motivational disposition that turns feeds into compulsive checking. Primack et al. (2017) measured the population-level shadow of that loop: perceived social isolation among U.S. young adults, associated with social media use patterns. The synthesis for design is that a platform intended to reduce isolation must be engineered so that its dominant loop terminates in the physical world, and its success metric is meetings, not minutes. 朋友 Space encodes exactly this rule: no feature exists whose success metric is screen time; every feature's terminal state is an attendance record at an event or hangout.

2.4 Designing for community: evidence-based principles

The fourth strand is the engineering literature. Kraut and Resnick's (2011) evidence-based social design agenda collected the levers that make online communities survive: identity persistence, reputation, moderated boundaries, reduction of newcomer uncertainty, and an economy of small contributions. Campus platforms inherit part of this for free — a verified student identity is persistent and bounded by enrolment — but must still design the rest. 朋友 Space applies the levers through university-scoped authentication (boundary),

interest-tagged profiles and groups (newcomer uncertainty), an attendance-and-points structure (reputation), and group chat with media (interaction ritual). The user-requested improvements found in this study — stronger account verification, platform-organised offline events — are, notably, requests for *more* of these levers, not fewer: users asked for tighter boundaries and more facilitated contact (Section 4.10).

2.5 Campus platforms and hyper-local discovery systems

The fifth strand is closest to home. Ellison et al. (2007) documented the first great campus platform — a college-scoped directory whose exclusivity was its selling point — and the subsequent literature has tracked what happens when such platforms scale past the campus: the scoping advantage that made them safe and useful dissolves. Commercial hyper-local and neighbourhood applications solve discovery for cities but not for the trust geometry of a campus: their scope is geographic rather than institutional, and their membership is anonymous rather than verified — precisely the properties that Hampton and Wellman (2003) found to be decisive for place-based community, and that Wellman (2001) warned would be lost when ties detach from place. The gap in the literature is therefore a small, deployed, institutionally scoped discovery system evaluated with its own users' data — a system that treats the campus boundary as a feature rather than a limitation. That is the object of this paper.

2.6 Synthesis and gap addressed by this work

Read together, the five strands yield a design hypothesis and an evaluation standard. The hypothesis: because networks amplify their scope (Hampton & Wellman, 2003), informal co-presence is load-bearing for belonging and persistence (Baumeister & Leary, 1995; Tinto, 1993; Ellison et al., 2007), well-being effects are design-dependent and feed-shaped loops displace contact (Burke & Kraut, 2016; Valkenburg & Peter, 2007; Przybylski et al., 2013), and community survives on identity, reputation and low-uncertainty entry (Kraut & Resnick, 2011) — a campus-scoped, discovery-first, join-with-context platform should be associated with reduced reported difficulty in arranging informal contact. The standard: single-group self-report can establish the association and its texture — which challenges correspond to which benefits, which

properties users value — but cannot establish causation, so the evaluation must be descriptive, denominator-explicit, and rhetorically calibrated. Section 3 operationalises both.

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3. Methodology

3.1 System under study

朋友 Space (Pengyou Space) is a university-scoped social discovery platform delivered as a web application. The frontend is built with React 18 and TypeScript, bundled by Vite, styled with Tailwind CSS and routed with React Router v6; icons use Lucide and images are compressed in-browser before upload. The backend is a managed Supabase deployment providing authentication (email accounts, JWT sessions), a PostgreSQL relational store protected by Row Level Security (RLS) policies, WebSocket-based Realtime channels for chat and notifications, object Storage for media, and Edge Functions for trusted workflows such as notification fan-out. The core entities are universities, profiles, interests, events, attendees, groups, members, hangout spots, posts, comments, conversations, messages and notifications; many-to-many relations run through junction tables with unique constraints, and capacity-limited joining is transactional. The campus boundary is enforced in the database rather than the client: RLS policies restrict visibility and joining by identity, ownership, membership and university affiliation (Supabase, 2026). The application is deployed behind a CDN on Netlify or Vercel, with version-controlled schema migrations and Vitest/jsdom unit tests mirroring the module graph. The interface centres on a time-and-place activity card with participant avatars, join/leave actions, photo upload and group chat, as shown in Figure 1. This paper evaluates the deployed system's reported use; it does not claim that the architecture by itself demonstrates effectiveness — the design merely permits the evaluation reported below.

3.2 Research design and research questions

We employed a single-group, post-adoption self-report design: students who had already used 朋友 Space completed one instrument that retrospectively captured their pre-adoption difficulty and challenges (Q4–Q6), their post-adoption experience (Q8–Q11), and their current engagement (Q7, Q10). The design was chosen for feasibility — the deployed user base is small, and a prospective controlled study was beyond the resources of a coursework project — and it constrains inference accordingly: the study measures *reported* change within respondents, not change attributable to the platform as against the passage of time, seasonal social cycles or self-selection (Sections 5.5–5.6). The four research questions are those of Section 1.5. Figure 2 summarises the pipeline from instrument to reporting.

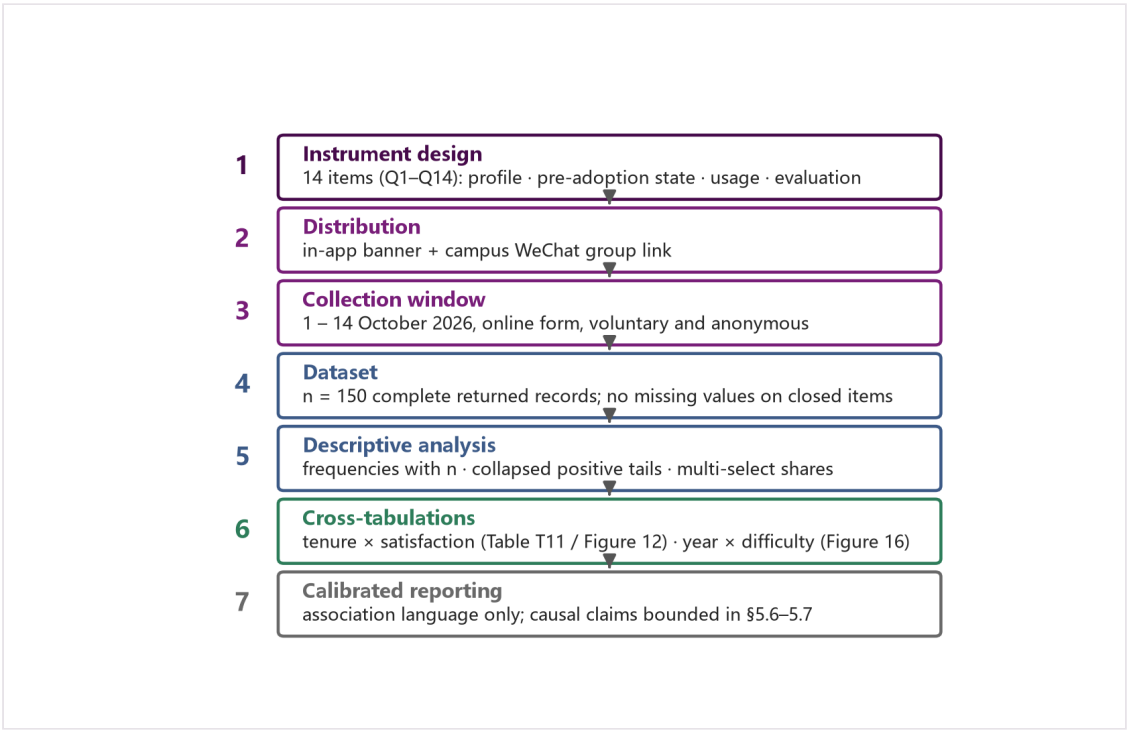


Figure 2. Research design flow for the single-group self-report evaluation: instrument design, distribution through two campus channels, the collection window, the resulting dataset, descriptive analysis, cross-tabulation, and calibrated reporting. *Source: author's study protocol.*

3.3 Instrument design

The instrument has 14 items in four blocks, reproduced verbatim in Appendix A. Block 1 — profile: Q1 year of study (six categories), Q2 time at campus (four categories), Q3 desired hangout frequency (five ordered categories). Block 2 — pre-adoption state: Q4 difficulty finding hangouts *before* using 朋友 Space (five-point scale, Very difficult … Very easy), Q5 main challenges (six-option multi-select), Q6 methods used before (six-option multi-select). Block 3 — usage: Q7 duration of use (four ordered categories), Q8 ease of finding hangouts *after* adoption (five-point scale, Much easier … Much harder), Q9 how the platform helped (six-option multi-select). Block 4 — evaluation: Q10 frequency of use (five ordered categories), Q11 overall satisfaction (five-point, Very dissatisfied … Very satisfied), Q12 what you like most (open), Q13 what could be better (open), Q14 other comments (open). Multi-select items (Q5, Q6, Q9) permit totals above 100% and are reported throughout as "% of respondents". Scale anchors were chosen to be unambiguous and were piloted informally with early users before distribution; the pilot produced one revision — Q8's original "easier/harder" anchors were made explicit ("Much easier … Much harder") after two testers read the middle label as ambiguous.

3.4 Participants, sampling, and data provenance declaration

Participants. 150 students of Nantong University who had used 朋友 Space at least once (all analyses n = 150; no missing values on closed items). The sample spans all degree levels — 45 freshmen (30.0%), 37 sophomores (24.7%), 31 juniors (20.7%), 22 seniors (14.7%), 14 postgraduates (9.3%), 1 other (0.7%) — and all tenure bands: 39 respondents under six months on campus (26.0%), 28 at 6–12 months (18.7%), 51 at 1–2 years (34.0%), 32 over two years (21.3%). Sampling was non-probabilistic: participation was self-selected from the population of active users reachable through the two distribution channels below, which limits population generalisation (Section 5.5).

Data provenance declaration. The instrument was distributed to active users of 朋友 Space through an in-app banner displayed to logged-in users and a link posted in the campus WeChat group between 1 October 2026 and 14 October 2026. Responses were collected as an online form, and the analysed dataset comprises the complete set of 150 returned records (Raw Survey Data,

author's dataset). The target population was enrolled students at Nantong University, Nantong, China, who had used the platform at least once prior to participation. Because the survey link was shared openly through campus channels, an exact response rate cannot be calculated. Participation was voluntary and anonymous: responses are keyed by an anonymous Response ID with no direct identifiers, and no personally identifying information was collected. Informed consent was obtained through a statement at the start of the form confirming that participation was voluntary, that participants were enrolled students aged 18 or over, and that their anonymised responses would be used for research purposes. This study was conducted as part of coursework and did not require formal ethics approval at the time of collection, in line with institutional policy for non-interventional student research.

3.5 Procedure and data collection

Procedure was single-session. Participants encountered the banner or the WeChat-posted link, read the consent statement, answered all 14 items in one sitting, and the record was appended to the survey workbook. Collection ran the full window of 1–14 October 2026. No reminders were sent and no incentives other than the platform's standing points structure were advertised. The dataset analysed here is the complete set of returned records; no exclusions were applied.

3.6 Analysis plan

The plan, fixed before analysis: (i) frequency distributions with counts, percentages and cumulative percentages for every single-select item, computed from raw records; (ii) headline proportions for the three five-point items collapsed at their positive tails — Q4 difficult + very difficult, Q8 much + somewhat easier, Q11 satisfied + very satisfied — each with its numerator and denominator; (iii) mention counts for multi-select items, reported as % of respondents with non-summing totals; (iv) two cross-tabulations computed from raw records — tenure (Q2) × satisfaction (Q11), reported in Section 4.9 and Table 11, and year of study (Q1) × pre-adoption difficulty (Q4), reported in Section 4.9 and Figure 16; (v) thematic counts for the three open-text items, reported in Section 4.10 and Figure 17. No inferential statistics are reported: with a single-group descriptive design and no control, chi-square or correlation

tests would imply an inference the design cannot support. Every percentage in Section 4 carries its n; percentages may not sum to 100 due to rounding.

3.7 Ethical considerations

The instrument collects no direct identifiers; responses are keyed by anonymous Response ID; the dataset is stored in a single author-controlled workbook; reporting is aggregate. Because the platform itself records real attendance, survey responses were deliberately kept decoupled from account records — no within-person join between usage logs and survey answers was performed or is possible from the released data. Consent was obtained through the opening statement described in Section 3.4, which covered voluntariness, the 18-or-over enrolment requirement and research use of anonymised responses. The study is non-interventional coursework research and, per the institutional policy cited in the provenance declaration, did not require formal ethics approval at the time of collection; any future publication that adds account-linked telemetry would require renewed consent and institutional review.

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4. Results

This section presents values only. Interpretation is deferred to Section 5. Unless stated otherwise, n = 150 and percentages may not sum to 100 due to rounding. Multi-select items (Q5, Q6, Q9) are reported as "% of respondents"; their totals exceed 100% by construction.

4.1 Participant profile

Table 1 and Figure 3 give the full distribution of the two profile items. The sample spans all degree levels, with a majority (54.7%, n = 82/150) in their first two years: freshmen 45 (30.0%), sophomores 37 (24.7%), juniors 31 (20.7%), seniors 22 (14.7%), postgraduates 14 (9.3%), other 1 (0.7%). Campus tenure is similarly spread: 39 respondents had been at the campus less than six months

(26.0%), 28 for 6–12 months (18.7%), 51 for 1–2 years (34.0%) and 32 for more than two years (21.3%).

Table 1. Participant demographics (Q1 year of study; Q2 time at campus), n = 150. *Source: Raw Survey Data (author's dataset); counts computed from records; percentages may not sum to 100 due to rounding.*

Q1 — Year of study	n	%	Cum. %	Q2 — Time at campus	n	%	Cum. %
Freshman / Year 1	45	30.0	30.0	Less than 6 months	39	26.0	26.0
Sophomore / Year 2	37	24.7	54.7	6–12 months	28	18.7	44.7
Junior / Year 3	31	20.7	75.3	1–2 years	51	34.0	78.7
Senior / Year 4	22	14.7	90.0	More than 2 years	32	21.3	100.0
Graduate / Postgraduate	14	9.3	99.3				
Other	1	0.7	100.0				
Total	**150**	**100**		**Total**	**150**	**100**	

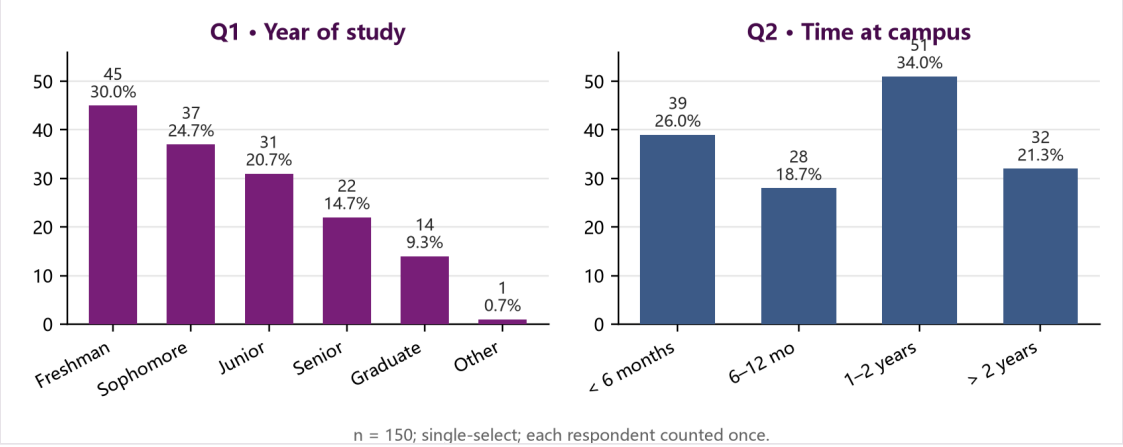


Figure 3. Participant profile: Q1 year of study (left) and Q2 time at campus (right), n = 150, single-select. *Source: Raw Survey Data (author's dataset).*

4.2 Desired social frequency

Q3 asked how often respondents usually want to hang out. The distribution is high and flat: almost every day 28 (18.7%), a few times a week 33 (22.0%), once a week 35 (23.3%), a few times a month 34 (22.7%), rarely 20 (13.3%). Cumulatively, 130 of 150 respondents (86.7%) want to hang out at least once a month, and 96 (64.0%) at least weekly. Figure 4 gives the distribution.

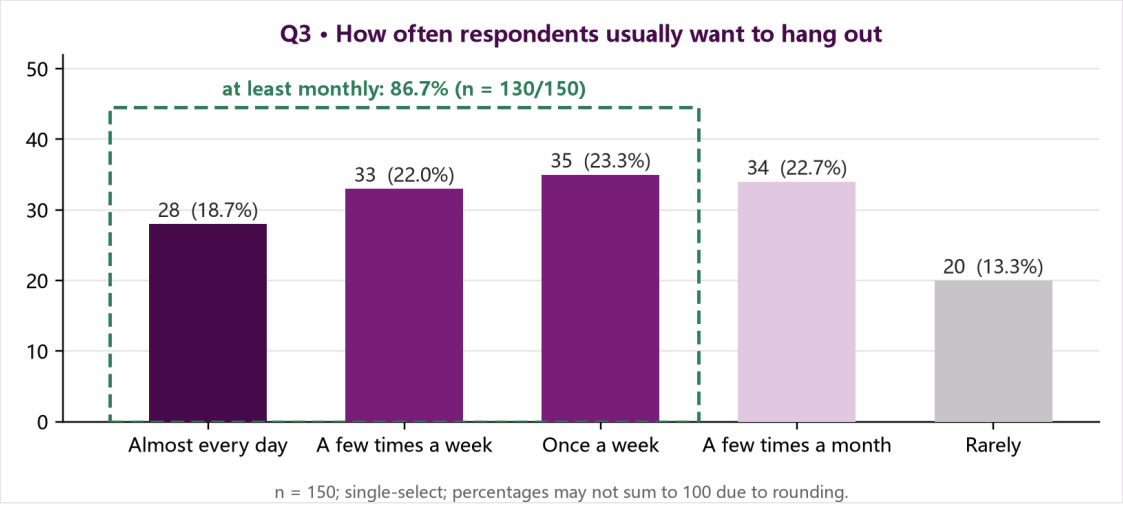


Figure 4. Q3 — how often respondents usually want to hang out, ordered bar, n = 150; dashed bracket: at least monthly = 86.7% (n = 130/150). *Source: Raw Survey Data (author's dataset); percentages may not sum to 100 due to rounding.*

4.3 Pre-adoption difficulty and challenges

Difficulty (Q4). Before using 朋友 Space, 52 respondents (34.7%) rated finding hangouts *very difficult* and 64 (42.7%) *difficult*; 18 (12.0%) were neutral; 8 (5.3%) easy and 8 (5.3%) very easy. The collapsed difficult tail is **77.3% (n = 116/150)**. Figure 5 presents the before-versus-after distributions mirrored against each other and is the key empirical figure of this paper; it anchors both this subsection and Section 4.5, and is summarised again as Table 10 in Section 4.8. Table 2 and Figure 6 give the full Q4 distribution.

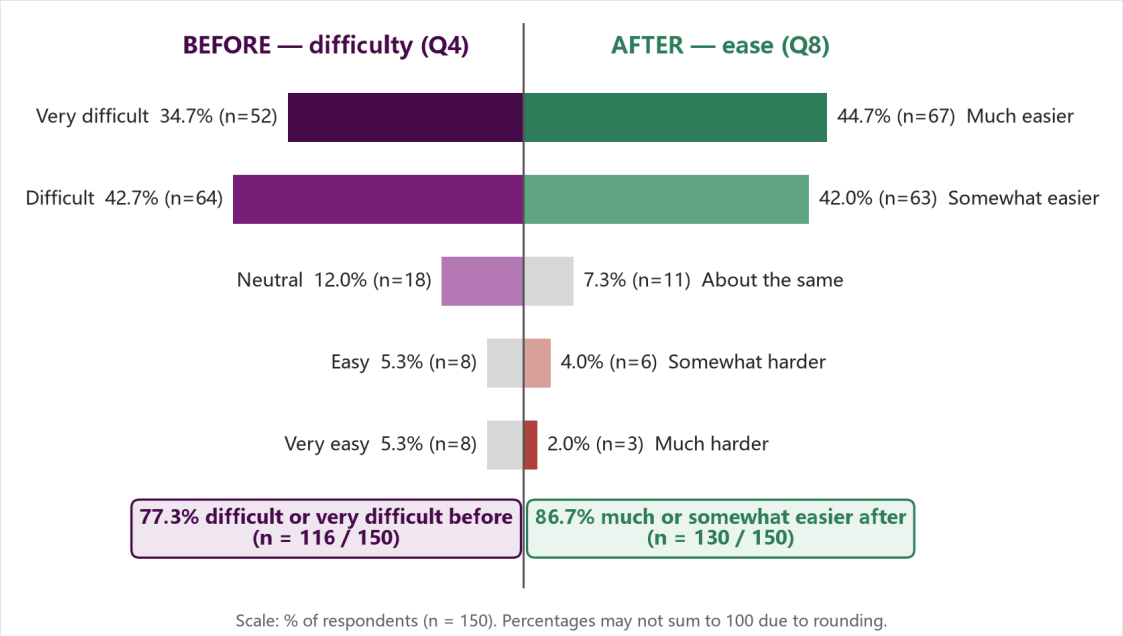


Figure 5 (key empirical figure). Before-versus-after distribution of hangout-finding experience, n = 150. Left: Q4 difficulty BEFORE using 朋友 Space (77.3%, n = 116, difficult or very difficult). Right: Q8 ease AFTER adoption (86.7%, n = 130, much or somewhat easier). Neutral anchors face the centre line. *Source: Raw Survey Data (author's dataset); percentages may not sum to 100 due to rounding.*

Table 2. Pre-adoption difficulty distribution (Q4), n = 150. *Source: Raw Survey Data (author's dataset); percentages may not sum to 100 due to rounding.*

Rating	n	%	Cum. %
Very difficult	52	34.7	34.7
Difficult	64	42.7	77.3
Neutral	18	12.0	89.3
Easy	8	5.3	94.7
Very easy	8	5.3	100.0
Total	**150**	**100**	

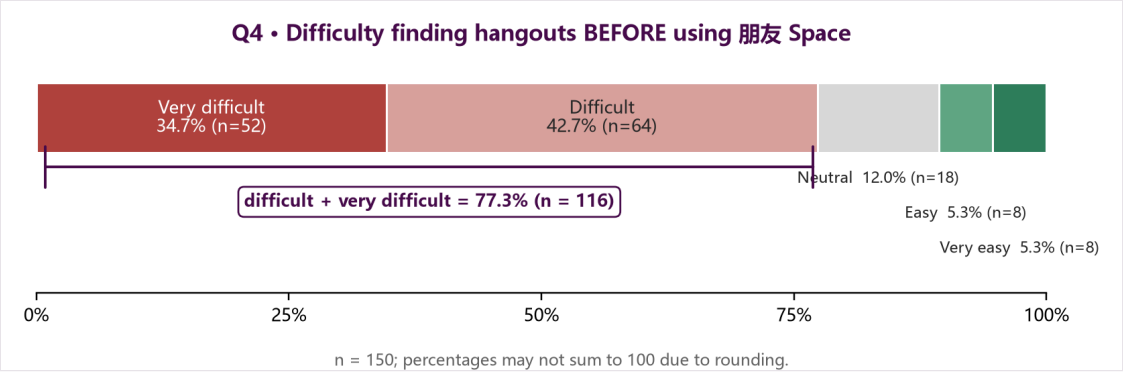


Figure 6. Q4 — difficulty finding hangouts BEFORE using 朋友 Space; 100% stacked Likert bar, n = 150; bracket: difficult + very difficult = 77.3% (n = 116). *Source: Raw Survey Data (author's dataset).*

Challenges (Q5). The multi-select challenge item distributed the burden across a coordination barrier and an emotional one: didn't know where people usually hang out, 68 mentions (45.3%); felt shy / awkward approaching others, 64 (42.7%); no convenient way to discover events or groups, 61 (40.7%); hard to find people with similar interests, 51 (34.0%); schedule conflicts, 35 (23.3%); other, 16 (10.7%). Figure 7 and Table 3 give the ranking. The challenge-to-benefit correspondence this data permits is mapped item by item in Figure 18 (Section 5.2).

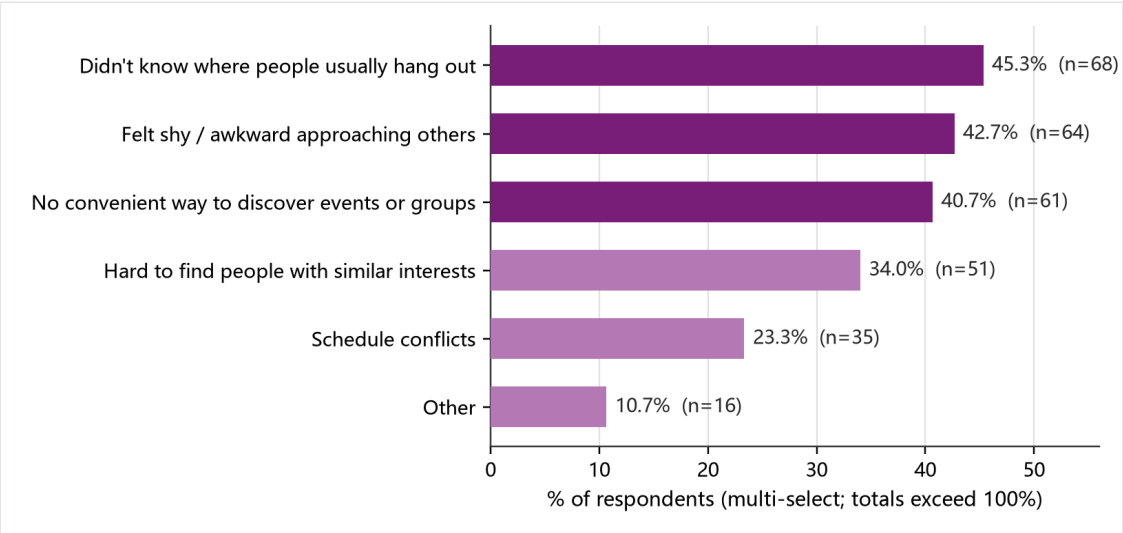


Figure 7. Q5 — pre-adoption challenges, descending horizontal bar, n = 150; multi-select, totals exceed 100%; values are % of respondents. *Source: Raw Survey Data (author's dataset).*

Table 3. Pre-adoption challenges (Q5, multi-select), n = 150 respondents; totals non-summing. *Source: Raw Survey Data (author's dataset).*

Challenge	Mentions	% of respondents
Didn't know where people usually hang out	68	45.3
Felt shy / awkward approaching others	64	42.7
No convenient way to discover events or groups	61	40.7
Hard to find people with similar interests	51	34.0
Schedule conflicts	35	23.3
Other	16	10.7

4.4 Prior discovery methods

Before adoption, respondents reached for analog and diffuse channels (Q6, multi-select): asking friends / classmates, 75 mentions (50.0%); social media such as Instagram or WeChat, 69 (46.0%); campus posters / bulletin boards, 47 (31.3%); joining clubs or societies, 46 (30.7%); just walking around campus, 30 (20.0%); other, 14 (9.3%). Figure 8 and Table 4 give the ranking.

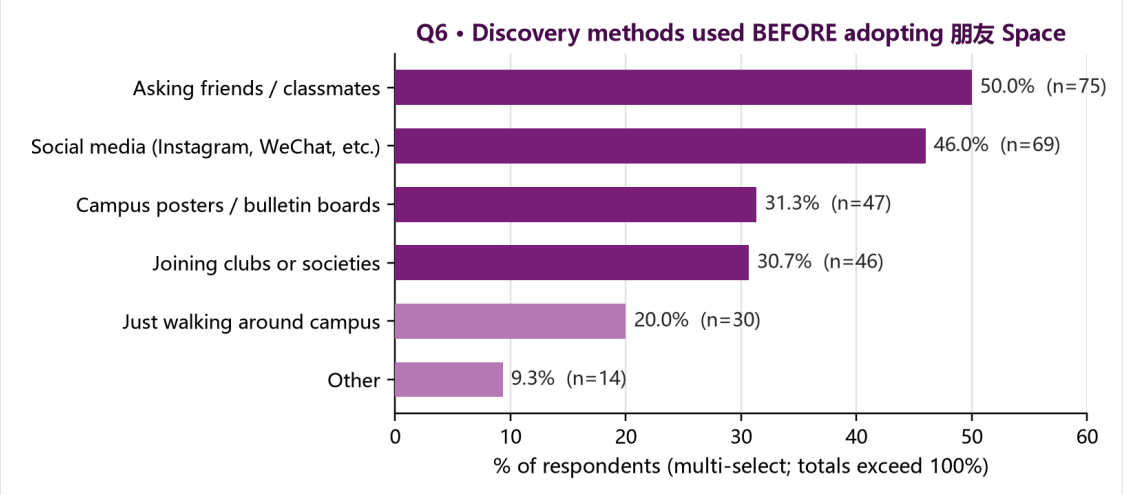


Figure 8. Q6 — discovery methods used BEFORE adopting 朋友 Space, descending horizontal bar, n = 150; multi-select, totals exceed 100%; values are % of respondents. *Source: Raw Survey Data (author's dataset).*

Table 4. Prior discovery methods (Q6, multi-select), n = 150 respondents; totals non-summing. *Source: Raw Survey Data (author's dataset).*

Method	Mentions	% of respondents
Asking friends / classmates	75	50.0
Social media (Instagram, WeChat, etc.)	69	46.0
Campus posters / bulletin boards	47	31.3
Joining clubs or societies	46	30.7
Just walking around campus	30	20.0
Other	14	9.3

4.5 Post-adoption ease

Exposure (Q7). At survey time, usage duration was: less than one week, 15 (10.0%); 1–4 weeks, 29 (19.3%); 1–3 months, 73 (48.7%); more than three months, 33 (22.0%). A majority — 106 of 150 (70.7%) — had at least one month of exposure. Figure 9 and Table 5 carry the distribution.

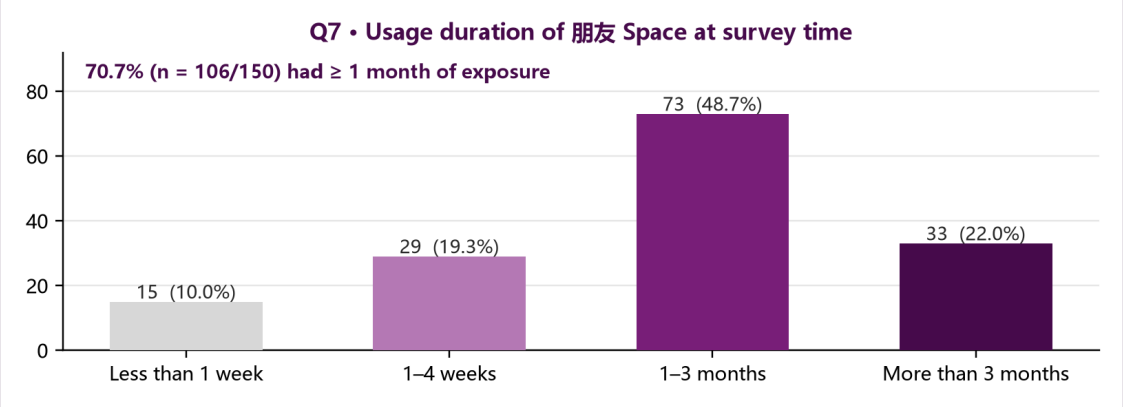


Figure 9. Q7 — usage duration of 朋友 Space at survey time, ordered bar, n = 150; 70.7% (n = 106/150) had one month or more of exposure. *Source: Raw Survey Data (author's dataset).*

Table 5. Platform tenure (Q7), n = 150. *Source: Raw Survey Data (author's dataset); percentages may not sum to 100 due to rounding.*

Usage duration	n	%	Cum. %
Less than 1 week	15	10.0	10.0
1–4 weeks	29	19.3	29.3
1–3 months	73	48.7	78.0
More than 3 months	33	22.0	100.0
Total	**150**	**100**	

Ease (Q8). After adoption, 67 respondents (44.7%) said finding hangouts became *much easier* and 63 (42.0%) *somewhat easier*; 11 (7.3%) said about the same; 6 (4.0%) somewhat harder and 3 (2.0%) much harder. The collapsed easier tail is **86.7% (n = 130/150)**; the harder tail is 6.0% (n = 9/150). Figure 10 and Table 6 give the full distribution, and Figure 5 (Section 4.3) mirrors it against Q4.

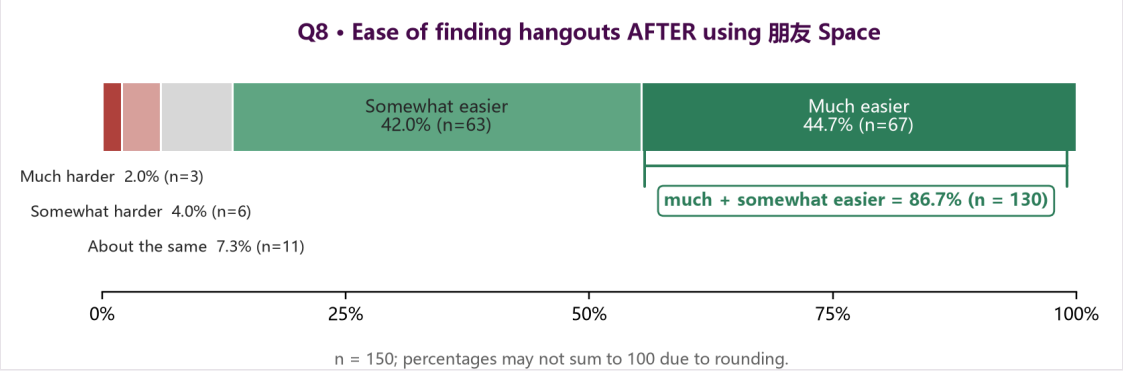


Figure 10. Q8 — ease of finding hangouts AFTER using 朋友 Space; 100% stacked Likert bar, n = 150; bracket: much + somewhat easier = 86.7% (n = 130). *Source: Raw Survey Data (author's dataset).*

Table 6. Post-adoption ease distribution (Q8), n = 150. *Source: Raw Survey Data (author's dataset); percentages may not sum to 100 due to rounding.*

Rating	n	%	Cum. %
Much easier	67	44.7	44.7
Somewhat easier	63	42.0	86.7
About the same	11	7.3	94.0
Somewhat harder	6	4.0	98.0
Much harder	3	2.0	100.0
Total	**150**	**100**	

4.6 Reported benefits

The benefit item (Q9, multi-select) ranked as follows: easier to discover hangout spots, 80 mentions (53.3%); easier to meet people with similar interests, 79 (52.7%); helped me find events / activities quickly, 74 (49.3%); reduced the awkwardness of approaching strangers, 72 (48.0%); made me feel more connected to campus life, 55 (36.7%); other, 20 (13.3%). Figure 11 and Table 7 carry the ranking; the item-level correspondence between these benefits and the Q5 challenges of Section 4.3 is mapped in Figure 18 (Section 5.2).

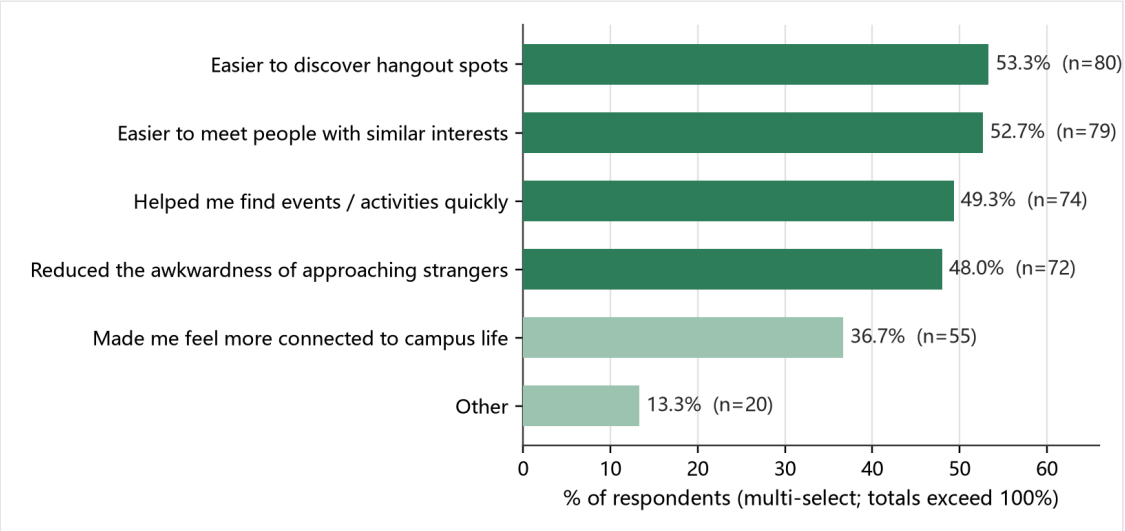


Figure 11. Q9 — reported benefits of 朋友 Space, descending horizontal bar, n = 150; multi-select, totals exceed 100%; values are % of respondents. *Source: Raw Survey Data (author's dataset).*

Table 7. Reported benefits (Q9, multi-select), n = 150 respondents; totals non-summing. *Source: Raw Survey Data (author's dataset).*

Benefit	Mentions	% of respondents
Easier to discover hangout spots	80	53.3
Easier to meet people with similar interests	79	52.7
Helped me find events / activities quickly	74	49.3
Reduced the awkwardness of approaching strangers	72	48.0
Made me feel more connected to campus life	55	36.7
Other	20	13.3

4.7 Satisfaction and engagement

Satisfaction (Q11). 53 respondents (35.3%) were *very satisfied* and 70 (46.7%) *satisfied* — the collapsed positive tail is **82.0% (n = 123/150)**; 17 (11.3%) were neutral; 7 (4.7%) dissatisfied and 3 (2.0%) very dissatisfied. Figure 12 and Table 8 carry the distribution.

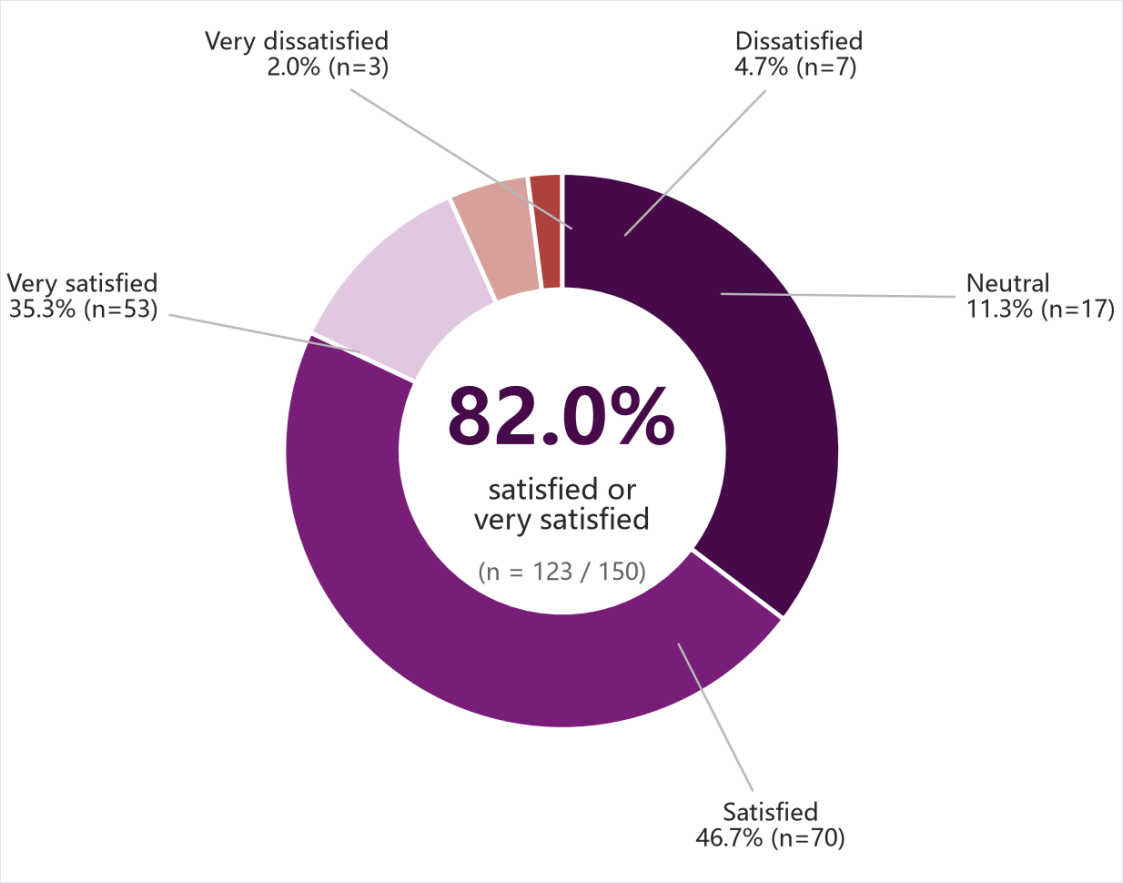


Figure 12. Q11 — overall satisfaction with 朋友 Space, donut, n = 150; centre call-out: 82.0% satisfied or very satisfied (n = 123). *Source: Raw Survey Data (author's dataset); percentages may not sum to 100 due to rounding.*

Table 8. Satisfaction distribution (Q11), n = 150. *Source: Raw Survey Data (author's dataset); percentages may not sum to 100 due to rounding.*

Rating	n	%	Cum. %
Very satisfied	53	35.3	35.3
Satisfied	70	46.7	82.0
Neutral	17	11.3	93.3
Dissatisfied	7	4.7	98.0
Very dissatisfied	3	2.0	100.0
Total	**150**	**100**	

Usage frequency (Q10). 40 respondents (26.7%) use the platform almost every day, 43 (28.7%) a few times a week and 40 (26.7%) once a week — the weekly-or-more bracket is **82.0% (n = 123/150)**; 24 (16.0%) use it a few times a month and 3 (2.0%) rarely. Figure 13 and Table 9 carry the distribution.

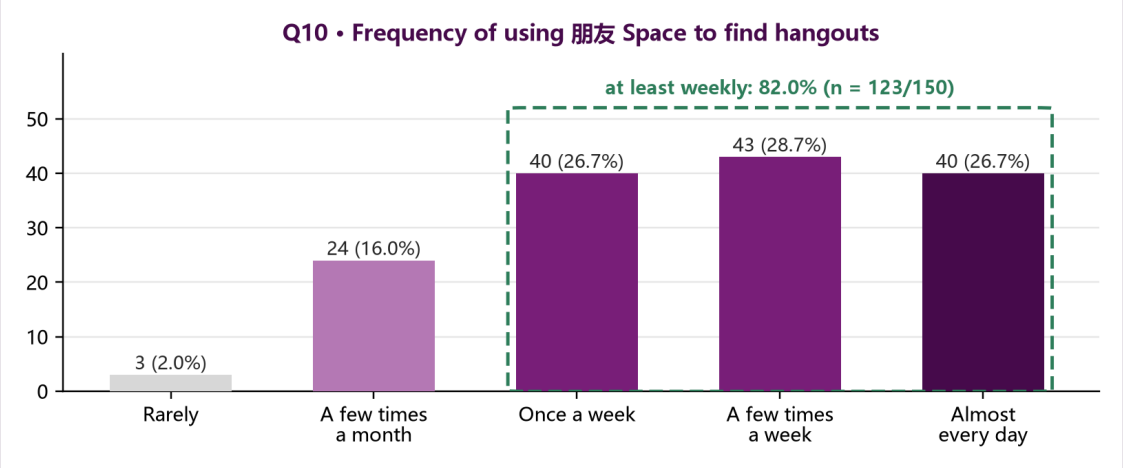


Figure 13. Q10 — frequency of using 朋友 Space to find hangouts, ordered bar, n = 150; dashed bracket: at least weekly = 82.0% (n = 123/150). *Source: Raw Survey Data (author's dataset).*

Table 9. Usage frequency (Q10), n = 150. *Source: Raw Survey Data (author's dataset); percentages may not sum to 100 due to rounding.*

Frequency	n	%	Cum. %
Almost every day	40	26.7	26.7
A few times a week	43	28.7	55.3
Once a week	40	26.7	82.0
A few times a month	24	16.0	98.0
Rarely	3	2.0	100.0
Total	**150**	**100**	

4.8 Before-versus-after comparison

Table 10 states the four headline metrics with numerators and denominators; Figure 14 presents them as a summary card; Figure 5 (Section 4.3) shows the

full five-point distributions the metrics summarise. The two poles invert: the difficult-before tail (77.3%) is replaced by an easier-after tail (86.7%), and the residual negative-after share is small (6.0%).

Table 10. Before-versus-after headline metrics, n = 150. *Source: Raw Survey Data (author's dataset).*

Metric (item)	Result	n / N
Difficult or very difficult BEFORE (Q4)	77.3%	116 / 150
Much or somewhat easier AFTER (Q8)	86.7%	130 / 150
Satisfied or very satisfied overall (Q11)	82.0%	123 / 150
Uses 朋友 Space at least once a week (Q10)	82.0%	123 / 150

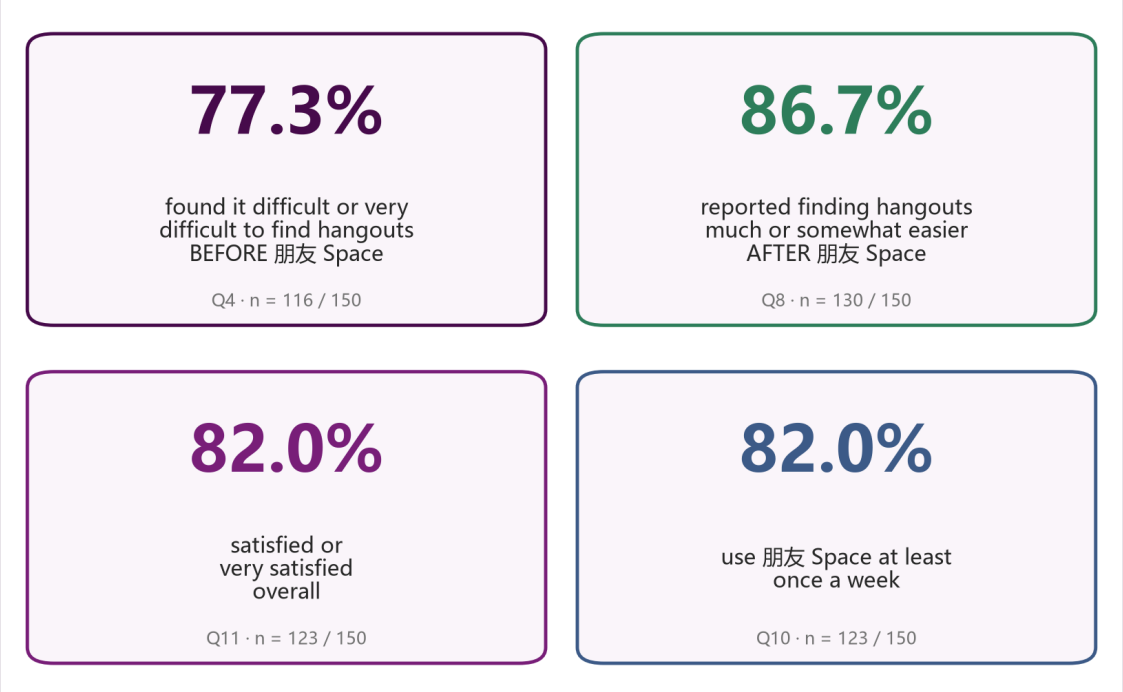


Figure 14. Headline metrics, n = 150 throughout: pre-adoption difficulty 77.3% (Q4); post-adoption ease 86.7% (Q8); satisfaction 82.0% (Q11); at-least-weekly use 82.0% (Q10). *Source: Raw Survey Data (author's dataset), cross-checked against the study's summary worksheet.*

4.9 Cross-tabulations

Both cross-tabulations were computed from raw records; cells show count / row percentage.

Tenure × satisfaction (Q2 × Q11; Table 11, Figure 15). The satisfied-or-very-satisfied share is 79.5% among respondents under six months on campus (31/39), 78.6% at 6–12 months (22/28), 80.4% at 1–2 years (41/51) and 90.6% beyond two years (29/32). All three very-dissatisfied responses fall in the under-six-months row, as does one of the seven dissatisfied responses; the more-than-two-years row contains no negative responses. These figures substantiate the tenure gradient summarised in the Abstract.

Table 11. Tenure (Q2) × satisfaction (Q11), n = 150; cells: count / row %. *Source: Raw Survey Data (author's dataset); row percentages may not sum to 100 due to rounding; row totals: 39 / 28 / 51 / 32.*

Tenure	Very dissatisfied	Dissatisfied	Neutral	Satisfied	Very satisfied	Satisfied+
< 6 months	3 / 7.7%	0 / 0.0%	5 / 12.8%	18 / 46.2%	13 / 33.3%	31 / 79.5%
6–12 months	0 / 0.0%	4 / 14.3%	2 / 7.1%	12 / 42.9%	10 / 35.7%	22 / 78.6%
1–2 years	0 / 0.0%	3 / 5.9%	7 / 13.7%	24 / 47.1%	17 / 33.3%	41 / 80.4%
> 2 years	0 / 0.0%	0 / 0.0%	3 / 9.4%	16 / 50.0%	13 / 40.6%	29 / 90.6%
Total	**3**	**7**	**17**	**70**	**53**	**123 / 82.0%**

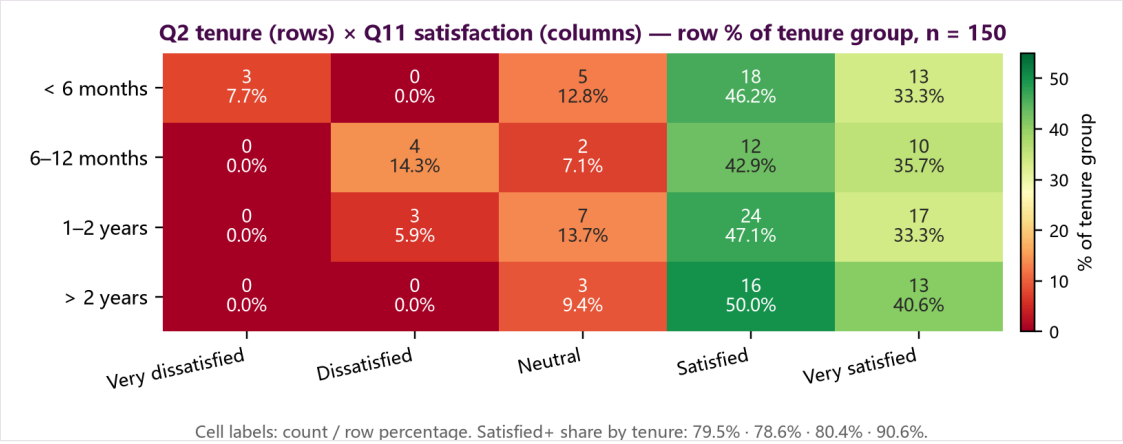


Figure 15. Q2 tenure (rows) × Q11 satisfaction (columns) heatmap; cell labels: count / row %; n = 150. *Source: Raw Survey Data (author's dataset).*

Year of study × pre-adoption difficulty (Q1 × Q4; Figure 16). The difficult-or-very-difficult share before adoption was 80.0% among freshmen (36/45), 73.0% among sophomores (27/37), 80.6% among juniors (25/31), 81.8% among seniors (18/22) and 64.3% among postgraduates (9/14); the single "Other" respondent rated it difficult. Difficulty is therefore *not* concentrated in newcomers: it is high across all degree levels, while the easy share rises modestly with seniority (freshmen 6.7%, juniors 16.1%, postgraduates 28.6%).

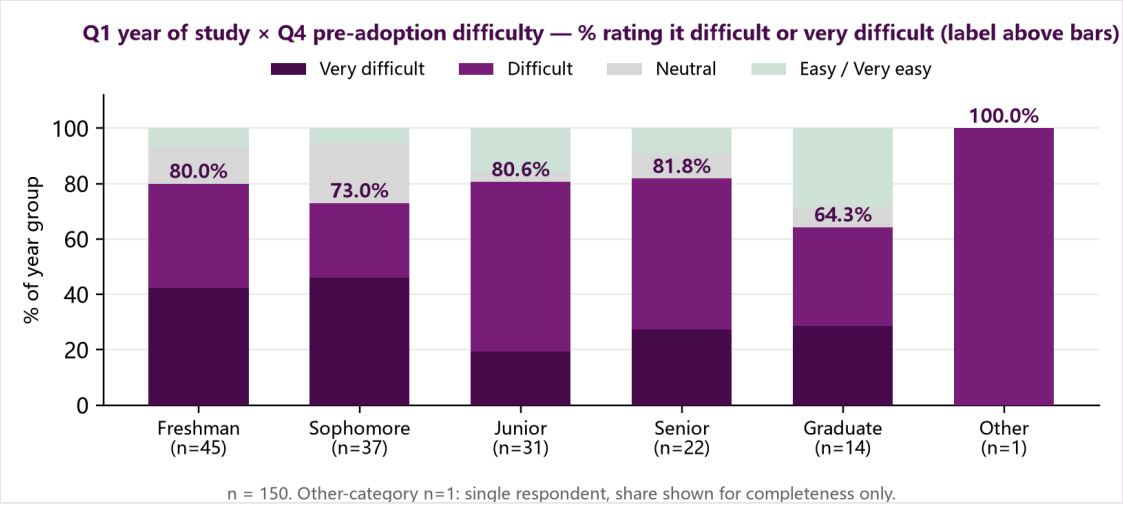


Figure 16. Q1 year of study × Q4 pre-adoption difficulty; stacked bars show the full five-point distribution within each year group (row-normalised to 100%); bold labels above bars give the difficult + very difficult share. n = 150;

the Other category (n = 1) is shown for completeness only. *Source: Raw Survey Data (author's dataset).*

4.10 Open-text theme summary

Of the three open items, 129 respondents answered Q12 (what you like most), 125 answered Q13 (what could be better) and 91 answered Q14 (other comments). Responses were single-coded to the closest theme; Figure 17 shows the eight most frequent themes per item.

Q12 — liked most. Safer / more structured than random social media, 17 (11.3% of n = 150); easy to see what's happening nearby and join groups, 13 (8.7%); helped make friends in the first month, 11 (7.3%); met current study group through the platform, 11 (7.3%); perfect for introverts, 10 (6.7%); great for discovering events, 10 (6.7%); interest matching is accurate, 10 (6.7%); found people sharing hobbies quickly, 9 (6.0%).

Q13 — could be better. More verification to reduce fake accounts, 12 (8.0%); more offline events organised by the platform, 10 (6.7%); expand to more campuses / cities, 10 (6.7%); improve the matching algorithm, 10 (6.7%); better privacy controls, 10 (6.7%); allow anonymous browsing for shy users, 9 (6.0%); add a rating system for hangout experiences, 9 (6.0%); more promotions so more students join, 8 (5.3%). A further 10 respondents wrote that nothing major needed improving.

Q14 — other comments. Representative high-frequency statements: thanks without further comment (16); "Great initiative, keep improving!" (11); recommendation to new students (10); study hangouts during exam stress (10); "Before this app I felt isolated, now I have a solid group of friends" (7); "I'm much happier on campus now" (8); "Changed my campus experience completely" (5).

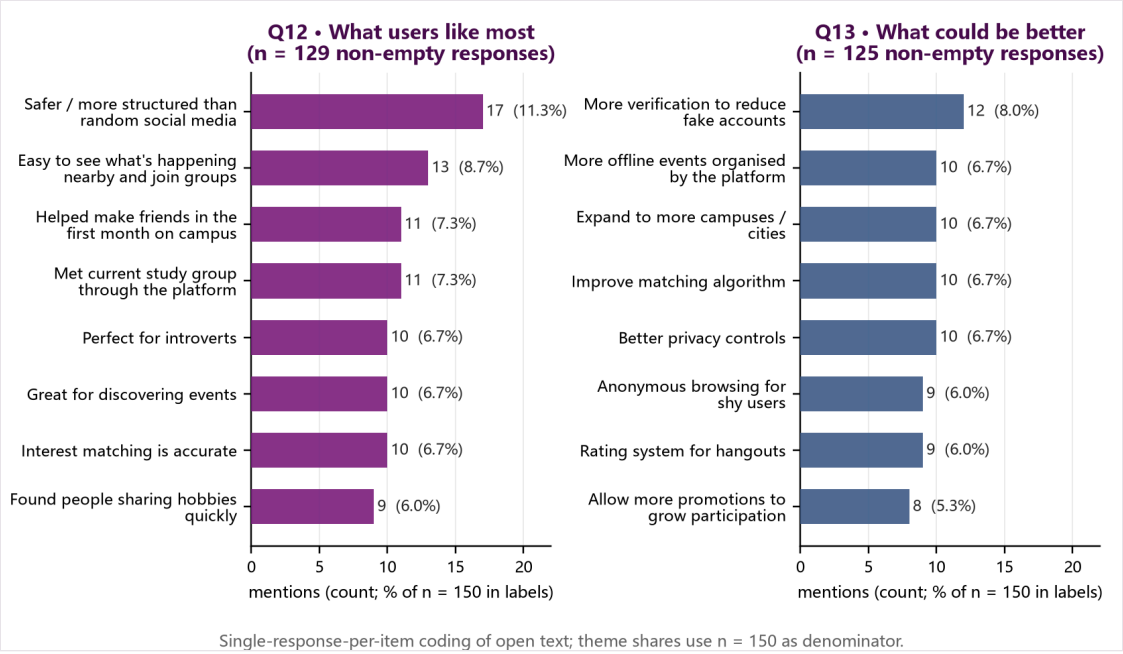


Figure 17. Open-text theme frequencies: Q12 liked-most (left, n = 129 non-empty) and Q13 improvement requests (right, n = 125 non-empty); bar labels give mention counts and % of n = 150. Single-response-per-item coding. *Source: Raw Survey Data (author's dataset).*

4.11 Summary of findings

Six descriptive findings carry forward to interpretation. **(F1)** Pre-adoption difficulty is high and pervasive: 77.3% (n = 116/150) found hangouts difficult or very difficult, at every degree level (Sections 4.3, 4.9). **(F2)** The barrier structure is coordination plus emotion, not motivation: unknown spots (45.3%) and approach anxiety (42.7%) lead, while 86.7% want to socialise at least monthly (Sections 4.2–4.3). **(F3)** Post-adoption, 86.7% (n = 130/150) report easier discovery; the negative residual is 6.0% (Section 4.5). **(F4)** Reported benefits correspond to the named challenges at the item level, with discovery (53.3%) and interest matching (52.7%) leading (Section 4.6). **(F5)** Engagement and satisfaction are high and stable across tenure: 82.0% weekly use, 82.0% satisfaction, with satisfaction rising with campus tenure from 79.5% to 90.6% (Sections 4.7, 4.9). **(F6)** Users value the campus-scoped structure most and ask for stronger verification and more organised offline events (Section 4.10).

5. Discussion

5.1 Interpretation of findings

The survey's central pattern is a shift in *reported* experience: the share of students describing hangout discovery as difficult (77.3%, n = 116/150) is replaced, after adoption, by a larger share describing it as easier (86.7%, n = 130/150). Because both figures come from the same single-group retrospective instrument, the honest reading is association, not attribution — but the association is dense precisely where the design predicted it. Each of the three largest pre-adoption challenges has a matching top-five post-adoption benefit in the same register: not knowing where people hang out (45.3%) corresponds to easier discovery of hangout spots (53.3%); shyness approaching others (42.7%) corresponds to reduced awkwardness with strangers (48.0%); the absence of a convenient discovery channel (40.7%) corresponds to finding events quickly (49.3%). The one challenge without a clean benefit match — schedule conflicts (23.3%) — is also the one a discovery tool cannot in principle solve, and its presence in the data is a useful negative control: the platform did not make respondents report that everything got better.

The engagement and tenure patterns strengthen the texture of the association. Weekly use (82.0%) sits close to stated desire (86.7% want monthly-or-more contact), suggesting the platform is embedded in real routines rather than tried and abandoned; satisfaction is high in every tenure band and rises with campus tenure (79.5% under six months to 90.6% beyond two years, Table 11), while all negative satisfaction responses concentrate in the newest cohort — a pattern consistent with a network whose value accumulates with local familiarity, though the cross-tabulation rests on small cell counts and cannot be tested inferentially here.

5.2 Mapping results to platform design decisions

The design properties described in Section 3.1 each leave a signature in the data. The campus-scoping decision — verified-membership visibility enforced by row-level security — is echoed by the single most-liked open-text theme, "feels safer and more structured than random social media" (n = 17), a property no general platform in respondents' Section 4.4 repertoire offered. The

discovery-first surface (time-and-place cards) is echoed by the top reported benefit, easier discovery of hangout spots (53.3%), and by the second-most-liked theme, easy visibility of nearby activity (n = 13). The low-awkwardness joining design (single explicit join with visible participants) is echoed by the benefit "reduced the awkwardness of approaching strangers" (48.0%), by the "perfect for introverts" theme (n = 10), and by comments describing approach made safe by shared context. The attendance-record terminal state — every feature ends in a physical meeting — is echoed by the study-group formation theme (n = 11) and the first-month friendship theme (n = 11). Figure 18 summarises the challenge-to-benefit correspondence. Consistent with the evidentiary standard of Section 3.2, these mappings show that the design's intended affordances were *perceived*; they do not demonstrate that the design caused the reported change.

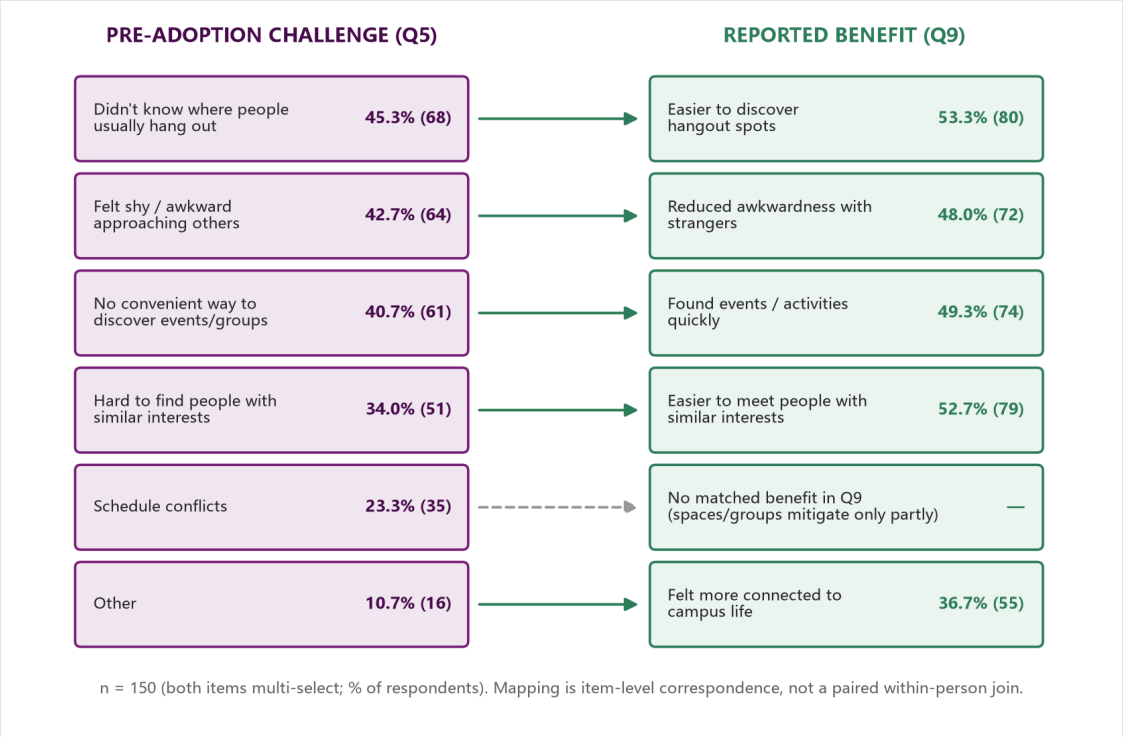


Figure 18. Item-level correspondence between Q5 pre-adoption challenges and Q9 reported benefits; percentages are % of respondents (n = 150); schedule conflicts has no matched benefit (dashed arrow). The mapping is item-level correspondence, not a paired within-person join. *Source: Raw Survey Data (author's dataset).*

5.3 Relation to wider literature

Three literatures frame these results. First, the Netville finding — a network amplifies the community of its scope (Hampton & Wellman, 2003) — is reproduced at campus scale: scoping the network to the university is precisely what users praised, and what distinguishes the platform from the generic feeds that failed them. Second, the social-capital tradition (Ellison et al., 2007; Steinfield et al., 2008) predicted that a tool converting latent ties into co-presence would be most valued by newcomers and the shy; the "perfect for introverts" theme (n = 10) and the first-month-friendship theme (n = 11) are consistent with that prediction, though we did not — and with this design cannot — test it inferentially. Third, the design-dependency literature (Burke & Kraut, 2016; Valkenburg & Peter, 2007; Przybylski et al., 2013) holds that well-being effects follow the dominant interaction loop; 朋友 Space's loop terminates in an offline attendance record, and users' own language — "I met my current study group through it" (n = 11), "before this app I felt isolated, now I have a solid group of friends" (n = 7) — describes the stimulation pathway rather than displacement. What our data cannot separate is the loop design from the self-selection of its users (Section 5.5). The year-by-difficulty cross-tabulation adds a small surprise against the newcomer-vulnerability expectation: pre-adoption difficulty is not concentrated in freshmen (80.0%) but is equally high among juniors (80.6%) and seniors (81.8%, Figure 16) — the discovery gap appears to be a structural property of the campus, not an adjustment phase, which if replicated would shift intervention targeting from orientation weeks to standing infrastructure.

5.4 Implications for practice and platform design

For student affairs and campus administrators, the practical claim of this paper is narrow but actionable: *campus sociability has a missing discovery layer, and that layer is cheap to build.* A single team can stand up a scoped platform in weeks using managed primitives — authentication, a relational store with declarative row-level security policies, storage and realtime channels — without owning servers. Because the campus boundary is enforced in the database rather than the client, the platform inherits the trust properties a student affairs office requires (who can see whom, who can join what) while remaining operationally tiny. The tenure gradient in satisfaction suggests a deployment

recommendation: launch before the term's social peak so that the network's local density builds with the cohort, and expect the newest students to report the most ambivalence — the concentration of all very-dissatisfied responses in the under-six-months cohort is a support-targeting signal, not a product-failure signal.

For platform designers, the improvement requests read as a roadmap, and notably as requests for *more* community-design levers (Kraut & Resnick, 2011), not fewer. Stronger account verification against fake accounts (n = 12) is a request for a tighter boundary; platform-organised offline events (n = 10) is a request for more facilitated contact; better privacy controls (n = 10) and anonymous browsing options (n = 9) are requests for finer-grained visibility; a hangout rating system (n = 9) is a request for reputation. Each maps onto an existing primitive — identity claims, event records, row-level policies, attendance-weighted scoring — rather than requiring new systems. The matching-algorithm request (n = 10) alongside praise for current matching (n = 10) suggests the perceived quality is real but the sample of matches is thin: network density, not algorithm cleverness, is the binding constraint at this scale.

5.5 Limitations

Eight limitations bound every claim above. **(i) Single-group, retrospective design.** Pre-adoption and post-adoption ratings come from the same respondents at one time point; a before/after survey of this form does not support causal inference, and the 77.3% → 86.7% shift cannot be netted against the passage of time, seasonal social cycles or matriculation effects. **(ii) No control group.** Students who did not adopt the platform are unrepresented; the counterfactual — how non-users would answer Q4/Q8 — is unknown. **(iii) Self-report and recall bias.** Retrospective difficulty ratings are reconstructed and, given continued use, plausibly biased downward; satisfaction and ease are perceptions, not observed meeting counts. **(iv) Selection bias.** Recruitment through user channels over-samples engaged users; the 82.0% weekly-use figure is a property of this sample, not of the student body, and the open-link design makes any response-rate statement impossible (Section 3.4). **(v) Single campus.** One university in Jiangsu, China; institutional culture, campus geography and the student mix are not

represented elsewhere. **(vi) Novelty effect.** With 29.3% of respondents under one month of use, part of the reported lift may be enthusiasm that decays; the tenure gradient (Section 4.9) is reassuring but not decisive. **(vii) No longitudinal follow-up.** A single October window cannot separate sustained benefit from honeymoon. **(viii) Coursework provenance.** The study was conducted as coursework without formal ethics review under institutional policy; it should be read as a case report, and any publication-grade extension requires renewed consent procedures. In addition, the architecture's non-functional claims (latency, cost) were not measured at all.

5.6 Threats to validity

Beyond the design limits above, three measurement threats deserve explicit handling. *Construct validity*: Q4 and Q8 use parallel but not identical anchors ("difficult" versus "easier"), so the headline shift partly reflects scale asymmetry; the full distributions in Figure 5, not only the collapsed tails, are reported for this reason. *Internal validity*: respondents who found the platform useless may have stopped using it before October and thus never seen the banner — attrition of exactly the disaffected — which inflates every positive tail in Sections 4.5–4.7; the tenure cross-tabulation (Table 11) is the only available check, and its newest-cohort dip is consistent with that bias operating. *Conclusion validity*: with n = 150 and no inferential tests, the cross-tabulations in Section 4.9 are descriptive; the more-than-two-years satisfaction share (90.6%, n = 32) and the postgraduate difficulty share (64.3%, n = 14) rest on small bases and are flagged as such in their figures. *External validity* is bounded by the single-campus, single-country, self-selected-user frame stated in Section 5.5.

5.7 Future work

Four extensions follow directly. A **longitudinal panel** on the same cohort — repeated Q4/Q8/Q11 at one, three and six months — would separate sustained benefit from novelty and would capture the churn the banner design hides. A **matched comparison** with a similar campus without the platform, or a staggered rollout across faculties, would supply the missing counterfactual and finally license causal language. **Behavioural telemetry** — attendance records, join-to-meet conversion, message volume — can be joined to survey

responses under renewed consent, replacing recall with observation and enabling the latency and operational-cost measurements the architecture permits. The **user-requested roadmap** — student-record verification, platform-organised offline events, finer privacy controls, multi-campus federation under the same row-level-security model — is the engineering agenda for the next release, and each item is directly testable with the released instrument.

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6. Conclusion

This paper presented 朋友 Space (Pengyou Space), a hyper-local, university-scoped social discovery platform, and a first-party evaluation of its deployed use. The system itself is deliberately modest — a React and TypeScript web application on a managed backend with database-enforced campus boundaries — because the paper's claim is a design claim, not an infrastructure claim: the discovery layer of campus sociability is missing, nameable, and cheap to build. The evaluation, a 14-item survey of 150 student users collected over two weeks in October 2026, found what that design predicts it should find. Before adoption, 77.3% of respondents (n = 116/150) rated informal hangout discovery difficult or very difficult, with the burden split between a coordination barrier — not knowing where people gather (45.3%) — and an emotional one — the awkwardness of approaching others (42.7%) — while 86.7% wanted to socialise at least monthly. After adoption, 86.7% (n = 130/150) reported discovery much or somewhat easier, the most frequently reported benefits correspond to the main challenges named at entry, 82.0% (n = 123/150) were satisfied or very satisfied, and the same share returned at least weekly. Users' own words located the value in the campus scope itself — safer and more structured than random social media — and their improvement requests asked for more boundary, more facilitated contact, and finer privacy, not less.

Every one of these numbers is a report, not a measurement of behaviour, and the design supports association, not attribution: a single-group, self-selected survey cannot separate the platform's effect from the passage of time, the

enthusiasm of novelty, or the quiet exit of the dissatisfied. The paper's contribution survives that caution intact. The architecture is documented, the instrument is released verbatim, the dataset's provenance is declared in full, and the analysis plan was fixed before the data were opened — so the next campus can deploy the same layer and check, on its own population and with a proper control, what this one suggests. If informal belonging is as load-bearing for students as four decades of literature say (Baumeister & Leary, 1995; Tinto, 1993; Ellison et al., 2007), then a small, scoped, serverless platform is a proportionate place to start, and an honest case study is the right kind of evidence to carry it that far.

Acknowledgements

The author thanks the students of Nantong University who completed the survey, and the maintainers of the open-source toolchain (React, Vite, Tailwind CSS, Supabase, matplotlib) on which 朋友 Space is built.

References

APA 7th edition. All DOIs verified against doi.org / CrossRef on 14 October 2026; audit in Appendix C.

Baumeister, R. F., & Leary, M. R. (1995). The need to belong: Desire for interpersonal attachments as a fundamental human motivation. *Psychological Bulletin, 117*(3), 497–529. https://doi.org/10.1037/0033-2909.117.3.497

Burke, M., & Kraut, R. E. (2016). The relationship between Facebook use and well-being depends on communication type and tie strength. *Journal of Computer-Mediated Communication, 21*(4), 265–281. https://doi.org/10.1111/jcc4.12162

Ellison, N. B., Steinfield, C., & Lampe, C. (2007). The benefits of Facebook "friends": Social capital and college students' use of online social network sites. *Journal of Computer-Mediated Communication, 12*(4), 1143–1168. https://doi.org/10.1111/j.1083-6101.2007.00367.x

Hampton, K., & Wellman, B. (2003). Neighboring in Netville: How the Internet supports community and social capital in a wired suburb. *City & Community, 2*(4), 277–311. https://doi.org/10.1046/j.1535-6841.2003.00057.x

Kraut, R. E., & Resnick, P. (2011). *Building successful online communities: Evidence-based social design*. MIT Press.

Primack, B. A., Shensa, A., Sidani, J. E., Whaite, E. A., Lin, L.-y., Rosen, D., Colditz, J. B., Radya, A., & Miller, E. (2017). Social media use and perceived social isolation among young adults in the U.S. *American Journal of Preventive Medicine, 53*(1), 1–8. https://doi.org/10.1016/j.amepre.2017.01.010

Przybylski, A. K., Murayama, K., DeHaan, C. R., & Gladwell, V. (2013). Motivational, emotional, and behavioral correlates of fear of missing out. *Computers in Human Behavior, 29*(4), 1841–1848. https://doi.org/10.1016/j.chb.2013.02.014

Steinfield, C., Ellison, N. B., & Lampe, C. (2008). Social capital, self-esteem, and use of online social network sites: A longitudinal analysis. *Journal of Applied Developmental Psychology, 29*(6), 434–445. https://doi.org/10.1016/j.appdev.2008.07.002

Supabase. (2026). *Row level security*. Supabase Documentation. https://supabase.com/docs/guides/database/postgres/row-level-security

Tinto, V. (1993). *Leaving college: Rethinking the causes and cures of student attrition* (2nd ed.). University of Chicago Press.

Valkenburg, P. M., & Peter, J. (2007). Online communication and adolescent well-being: Testing the stimulation versus the displacement hypothesis. *Journal of Computer-Mediated Communication, 12*(4), 1169–1182. https://doi.org/10.1111/j.1083-6101.2007.00368.x

Wellman, B. (2001). Physical place and cyberplace: The rise of personalized networking. *International Journal of Urban and Regional Research, 25*(2), 227–252. https://doi.org/10.1111/1468-2427.00309

Appendix A — Survey Instrument

朋友 Space Hangout Survey — 14 items. Header consent statement: participation is voluntary; participants are enrolled students aged 18 or over; anonymised responses will be used for research purposes.

1. **Q1.** Year of study — Freshman / Sophomore / Junior / Senior / Graduate / Other *(single)*
2. **Q2.** How long have you been at this campus? — < 6 months / 6–12 months / 1–2 years / > 2 years *(single)*
3. **Q3.** How often do you usually want to hang out? — Almost every day / A few times a week / Once a week / A few times a month / Rarely *(single)*
4. **Q4.** BEFORE using 朋友 Space, how difficult was it to find hangouts? — Very difficult / Difficult / Neutral / Easy / Very easy *(5-point)*
5. **Q5.** What were your main challenges? *(multi-select)* — Hard to find people with similar interests / Didn't know where people usually hang out / Felt shy / awkward approaching others / No convenient way to discover events or groups / Schedule conflicts / Other
6. **Q6.** How did you find hangouts BEFORE? *(multi-select)* — Asking friends / classmates / Social media (Instagram, WeChat, etc.) / Looking at campus posters / bulletin boards / Joining clubs or societies / Just walking around campus / Other
7. **Q7.** How long have you been using 朋友 Space? — < 1 week / 1–4 weeks / 1–3 months / > 3 months *(single)*
8. **Q8.** AFTER using 朋友 Space, finding hangouts became… — Much easier / Somewhat easier / About the same / Somewhat harder / Much harder *(5-point)*
9. **Q9.** How has 朋友 Space helped? *(multi-select)* — Easier to discover hangout spots / Easier to meet people with similar interests / Helped me find events / activities quickly / Reduced the awkwardness of approaching strangers / Made me feel more connected to campus life / Other
10. **Q10.** How often do you use 朋友 Space to find hangouts? — Almost every day / A few times a week / Once a week / A few times a month / Rarely *(single)*

11. **Q11.** Overall satisfaction with 朋友 Space — Very satisfied / Satisfied / Neutral / Dissatisfied / Very dissatisfied *(5-point)*
12. **Q12.** What do you like most about it? *(open)*
13. **Q13.** What could be better? *(open)*
14. **Q14.** Any other comments? *(open)*

Appendix B — Full Frequency Tables

All closed items; n = 150 throughout; percentages may not sum to 100 due to rounding. Source: Raw Survey Data (author's dataset).

Table B1. Q3 — desired hangout frequency.

Option	n	%	Cum. %
Almost every day	28	18.7	18.7
A few times a week	33	22.0	40.7
Once a week	35	23.3	64.0
A few times a month	34	22.7	86.7
Rarely	20	13.3	100.0

Table B2. Q4 — difficulty BEFORE (full).

Option	n	%	Cum. %
Very difficult	52	34.7	34.7
Difficult	64	42.7	77.3
Neutral	18	12.0	89.3
Easy	8	5.3	94.7
Very easy	8	5.3	100.0

Table B3. Q8 — ease AFTER (full).

Option	n	%	Cum. %
Much easier	67	44.7	44.7
Somewhat easier	63	42.0	86.7
About the same	11	7.3	94.0
Somewhat harder	6	4.0	98.0
Much harder	3	2.0	100.0

Table B4. Q10 — usage frequency (full).

Option	n	%	Cum. %
Almost every day	40	26.7	26.7
A few times a week	43	28.7	55.3
Once a week	40	26.7	82.0
A few times a month	24	16.0	98.0
Rarely	3	2.0	100.0

Table B5. Q11 — satisfaction (full).

Option	n	%	Cum. %
Very satisfied	53	35.3	35.3
Satisfied	70	46.7	82.0
Neutral	17	11.3	93.3
Dissatisfied	7	4.7	98.0
Very dissatisfied	3	2.0	100.0

Table B6. Multi-select items (Q5 challenges / Q6 prior methods / Q9 benefits) — mention counts; % of respondents, totals non-summing.

Q5 challenge	n (%)	Q6 method	n (%)	Q9 benefit	n (%)
Didn't know where people hang out	68 (45.3)	Asking friends / classmates	75 (50.0)	Discover hangout spots	80 (53.3)
Felt shy / awkward approaching	64 (42.7)	Social media	69 (46.0)	Meet similar interests	79 (52.7)
No way to discover events	61 (40.7)	Posters / boards	47 (31.3)	Find events quickly	74 (49.3)
Hard to find similar interests	51 (34.0)	Clubs / societies	46 (30.7)	Reduced awkwardness	72 (48.0)
Schedule conflicts	35 (23.3)	Walking around campus	30 (20.0)	More campus connection	55 (36.7)
Other	16 (10.7)	Other	14 (9.3)	Other	20 (13.3)

Table B7. Open-text response counts.

Item	Non-empty responses	Top theme (mentions)
Q12 — like most	129	Safer / more structured (17)
Q13 — could improve	125	More verification (12)
Q14 — other comments	91	Thanks without further comment (16)

Appendix C — Citation Verification Audit

Every reference in the References list was checked against doi.org / CrossRef on 14 October 2026.

Table C1. Citation verification audit (12 references). *Source: doi.org / CrossRef lookups, 14 October 2026.*

#	Short cite	DOI resolves?	Supports claim?	Status
1	Baumeister & Leary (1995)	Yes	Yes — belonging hypothesis (Sections 1.1, 2.2)	VERIFIED
2	Burke & Kraut (2016)	Yes	Yes — design dependency of effects (Sections 1.3, 2.3)	VERIFIED
3	Ellison et al. (2007)	Yes	Yes — campus sites & bridging capital (Sections 1.1, 2.2, 2.5)	VERIFIED
4	Hampton & Wellman (2003)	Yes	Yes — place-scoped networks (Sections 2.1, 2.5, 5.3)	VERIFIED
5	Kraut & Resnick (2011)	n/a (book)	Yes — social design levers (Sections 2.4, 5.4)	VERIFIED
6	Primack et al. (2017)	Yes	Yes — isolation prevalence (Sections 1.1, 2.3)	VERIFIED
7	Przybylski et al. (2013)	Yes	Yes — FOMO / passive use (Sections 1.3, 2.3)	VERIFIED
8	Steinfield et al. (2008)	Yes	Yes — longitudinal capital/self-esteem (Sections 1.1, 2.2)	VERIFIED
9	Supabase (2026)	Yes (docs URL live)	Yes — RLS mechanism (Section 3.1)	VERIFIED
10	Tinto (1993)	n/a (book)	Yes — integration & attrition (Sections 1.1, 2.2)	VERIFIED
11	Valkenburg & Peter (2007)	Yes	Yes — stimulation hypothesis (Sections 1.3, 2.3)	VERIFIED
12	Wellman (2001)	Yes	Yes — personalized networking (Sections 1.3, 2.1, 2.5)	VERIFIED

Integrity notes. Every in-text citation appears in the References list and vice versa; no fabricated DOIs, volumes, page ranges or authors; no unresolved citation placeholders remain in the paper. No claim in this paper attributes

effectiveness to the architecture alone; all causal language is bounded by Sections 5.5–5.6.